Combinatorics (Additive, extremal, graph, yayayayayay)

Raymond Xu

7 november 2026

Inhoudsopgave

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Yeah so a lot of combinatorics is about finding valid constructions. Actually this is a trend in all of math, so a background in what a construction is is great.

1 Set Theory

2 Extremal Combinatorics

Definition

Sunflower A sunflower is a collection of sets $S_1, S_2, \ldots, S_k$ such that $S_i \cap S_j$ is constant over all i, j.

It's called a sunflower because it looks like one when you draw it:

Theorem

Erdos-Rado Sunflower Lemma Given a family $\mathcal{F}$ of size s sets, we may find a sunflower of size k if $|\mathcal{F}| > s!(k-1)k^{s-1}$

Proof

Define $f(s,k)$ to be the maximal cardinality of $\mathcal{F}$ such that it does not contain a sunflower. Continue with induction on s, and suppose that no sunflower exists in $\mathcal{F}$. If $s = 1$, clearly $f(1,k) \leq k-1$. To induct $s-1 \to s$, let $A_1, \ldots, A_t$ be the maximum collection of pairwise disjoint sets in $\mathcal{F}$. Then define $A = \cup_{1 \leq i \leq t} A_i$ so that $|A| < sk$, or else we can select k disjoint sets. Define $\mathcal{F}_a = \{S \not\ni a : a \in S, S \in \mathcal{F}\}$. Bound $|\mathcal{F}| \leq \sum_a |\mathcal{F}_a| \leq f(s-1,k) \cdot sk \leq s!(k-1)k^{s-1}$ as desired.

Problem

3 Standard Olympiad Combinatorics

Problem

2025 IMO 6 Consider a 2025×2025 grid of unit squares. Matilda wishes to place on the grid some rectangular tiles, possibly of different sizes, such that each side of every tile lies on a grid line and every unit square is covered by at most one tile.

Determine the minimum number of tiles Matilda needs to place so that each row and each column of the grid has exactly one unit square that is not covered by any tile.

Solution

Color all uncovered squares in the $n^2 \times n^2$ grid. Define a path to be a sequence of squares $s_1, \ldots, s_k$ such that s_{i+1} has a larger x and y coordinate than s_i. We can always partition the grid into paths $\mathcal{P}_1, \ldots, \mathcal{P}_j$ such that every square is included in exactly 1 path.

Claim: The minimum number of rectangles needed to satisfy the condition is exactly

$$\sum_{i=1}^{j} |P_i| + 2(n-1) - 1 = n^2 + 2n - 3$$

Let's generalize this.

Problem

Buffed to 3D

3.1 Invariant

3.2 Bounding

4 Graphs

Lemma

Lindström–Gessel–Viennot Let G be a weighted, acyclic graph, and let $A = \{a_1, a_2, \ldots, a_n\}$ and $B = \{b_1, \ldots, b_n\}$ be starting and ending vertices.

Lemma

Suppose that for any $i < j$ and $k < l$, every path $P_1 : a_i \to b_l$ intersects every path $P_2 : a_j \to b_k$. For two vertices, let $w(a, b)$ be the weight over the product of edges from $a \to b$. If

$$M = \begin{pmatrix} w(a_1, b_1) & w(a_1, b_2) & \dots & w(a_1, b_n) \\ w(a_2, b_1) & w(a_2, b_2) & \dots & w(a_2, b_n) \\ \vdots & \vdots & \ddots & \vdots \\ w(a_n, b_1) & w(a_n, b_2) & \dots & w(a_n, b_n) \end{pmatrix}$$

Let $P_1, \dots, P_n$ be an n-tuple of paths from A to B, and let σ be a permutation of $[n]$. Then

$$\det(M) = \sum_{(P_1,\dots,P_n):A\to B} \operatorname{sign}(\sigma(P)) \prod_{i=1}^{n} w(P_i)$$

where $\operatorname{sign}(\sigma) = (-1)^{N(\sigma)}$ where $N(\sigma)$ is the number of inversions.

Proof

By the definition of M,

$$\det(M) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) \prod_{i=1}^{n} w(a_i, b_{\sigma(i)})$$

$$= \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) \sum \{w(P) : P \text{ is an } n\text{-tuple of paths } \}$$

$$= \sum_{P=(P_1,\dots,P_n):A\to B} \operatorname{sign}(\sigma(P)) w(P)$$

Problem

2024 Putnam A6 Let c_0, c_1, c_2, ... be a sequence defined so that

$$\frac{1 - 3x - \sqrt{1 - 14x + 9x^2}}{4} = \sum_{k=0}^{\infty} c_k x^k$$

for sufficiently small x. For a positive integer n, let A be the n-by-n matrix with i, j-entry c_{i+j-1} for i and j in $\{1, \dots, n\}$. Find the determinant of A.

Solution

Solution 1 Weight every edge of the cartesian plane $(a, b) \to (a+1, b)$ with 5 if $b \equiv a \pmod 2$ and 2 if $b \equiv a + 1 \pmod 2$.
Claim: The sum of the paths from $(0, 0)$ to $(n-1, n-1)$ is exactly c_n.
Proof. Let d_i be the sum of the weights with the 2's and 5's reversed. Then

$$c_n = 5(c_{n-1}d_1 + c_{n-2}d_2 + \ldots + c_1 d_{n-1})$$

$$d_n = 2(c_{n-1}d_1 + \ldots + c_1 d_{n-1})$$

so $d_n = \frac{2}{5}c_n$ for $n \geq 2$, so

$$c_n = 2(c_1 c_{n-1} + \ldots + c_{n-1}c_1) + 3c_{n-1}$$

Let $a_i = (n - i + 1, n - i + 1)$ and $b_i = (n + i - 1, n + i - 1)$ so applying LGV gives $10^{\binom{n}{2}}$ as the answer.

Solution

Solution 2

5 Additive Combinatorics